FAST GRAPH-BASED ALGORITHMS FOR IMAGE SEGMENTATION

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OVERVIEW

- Introduction to Graph Theory
- 2-Way Graph Partitioning
- Algebraic Multigrid Method (with Heavy Edge Coarsening)
- Applications: Image Segmentation

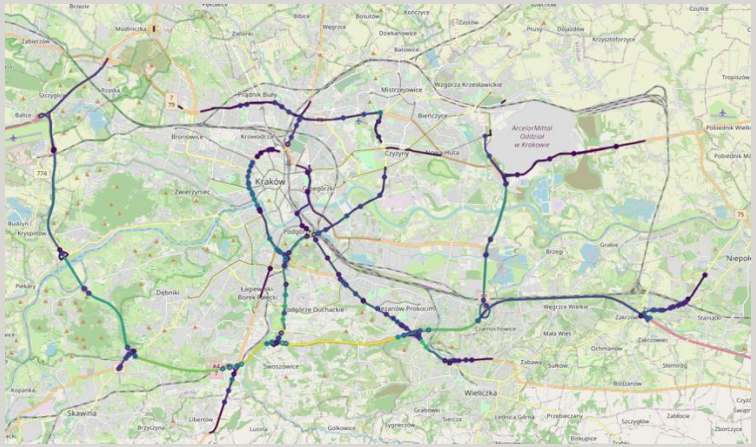
INTRODUCTION TO GRAPH THEORY

INTRODUCTION TO GRAPH THEORY – SOCIAL NETWORK ANALYSIS

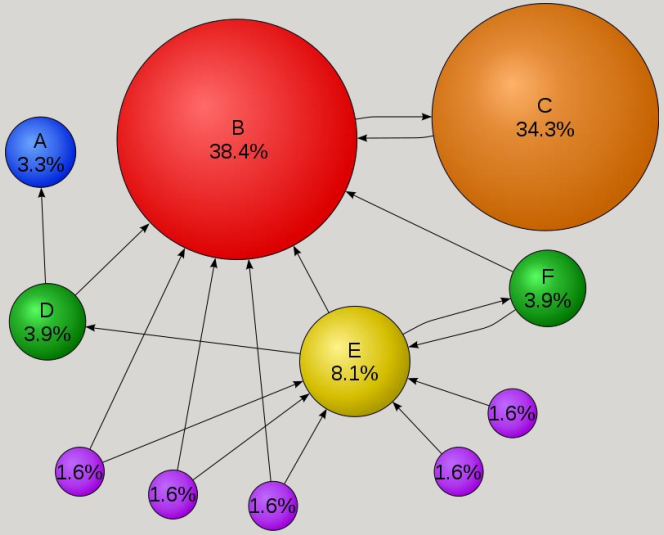
INTRODUCTION TO GRAPH THEORY – MORE EXAMPLES

Neural Network

Transportation Network



Google's PageRank Algorithm



Protein-Protein Interactions (PPIs)

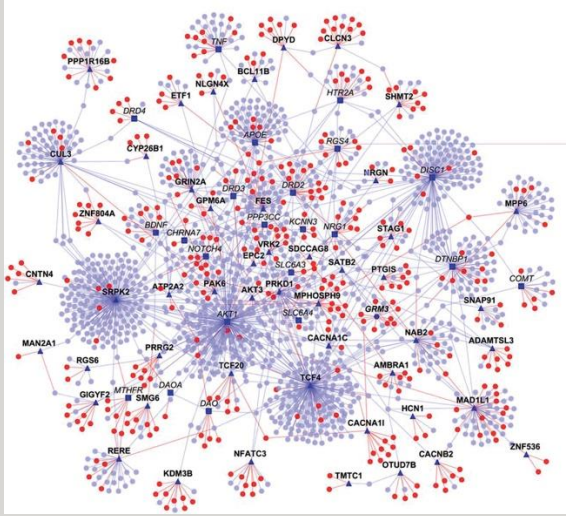
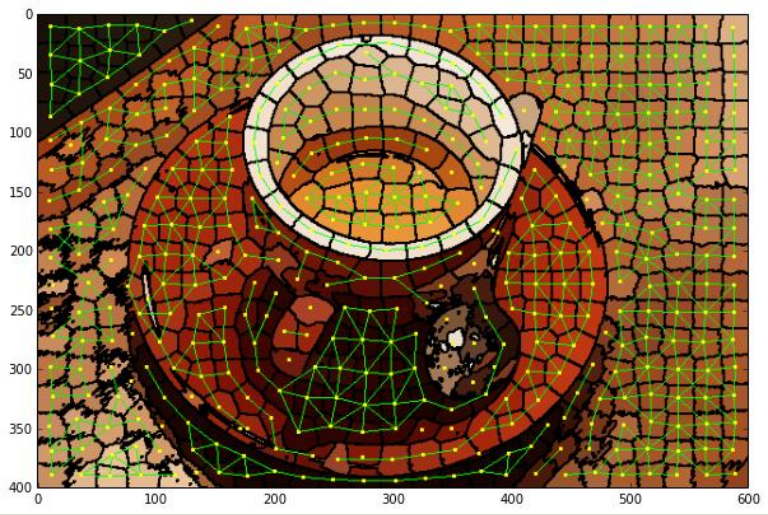


Image Segmentation



Graph theory is the study of graphs – mathematical structures used to model pairwise relations between objects.

Spectral graph theory is the study of the properties of a graph in relation to the characteristic polynomial, eigenvalues, and eigenvectors of matrices associated with the graph, particularly the Laplacian matrix.

NOTATION

Suppose $G = (V, E, W)$ consists of three finite sets:

Vertex set V,

$$V \neq \emptyset$$
$$v_i \in V$$

Edge set E,

$$e_{ij} = (v_i, v_j) \in E$$

and Weight set W

$$w_{ij} \in W$$

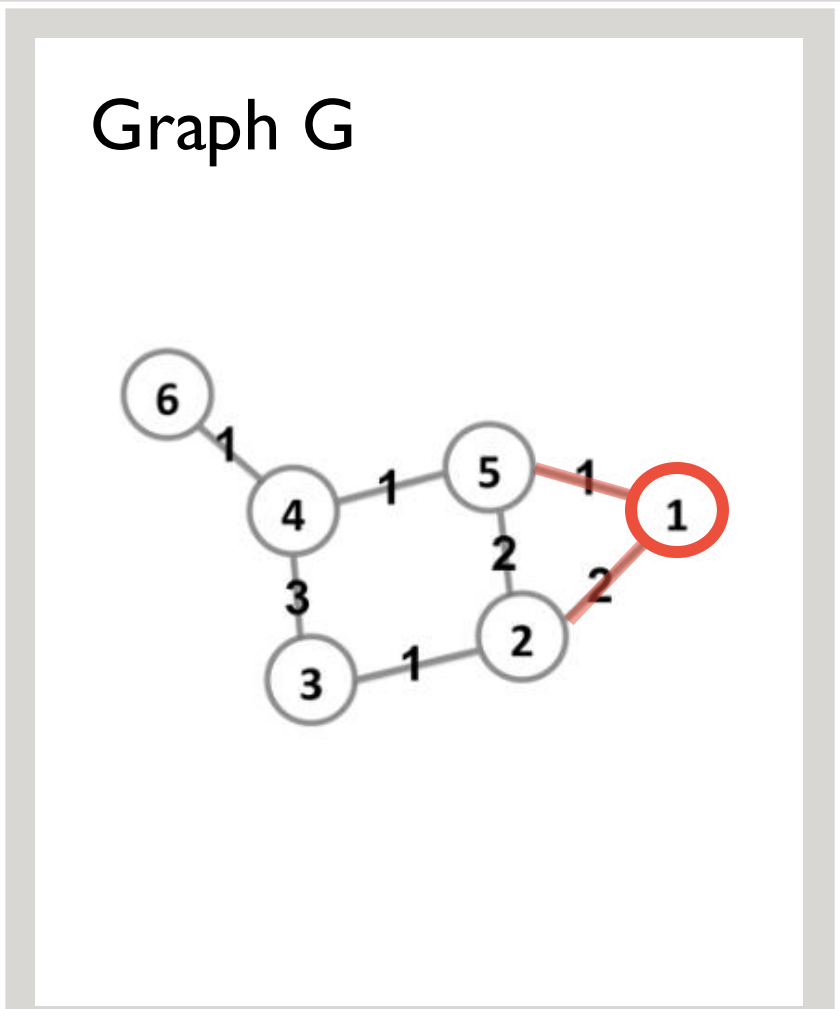
Degree $d_i = d(v_i)$: the number of edges connected to v_i

Weighted Degree $\delta_i = \delta(i) = \sum w_{ij}$

<u>Example</u>: v_1

Degree $d_1 = d(v_1) = 2$

Weighted Degree $\delta_1 = \delta(1) = 1 + 2 = 3$



ADJACENCY MATRIX

Suppose $G = (V, E, W)$ is a connected undirected graph.

Unweighted Adjacency Matrix *A*:

$$A_{ij} = \begin{cases} 1, & \text{if } e_{ij} \in E \\ 0, & \text{otherwise} \end{cases}$$

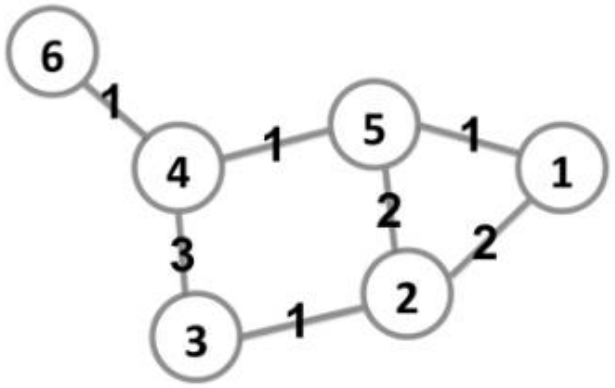
Weighted Adjacency Matrix *W*:

$$W_{ij} = \begin{cases} w_{ij}, & \text{if } e_{ij} \in E \\ 0, & \text{otherwise} \end{cases}$$

Adjacency Matrix for Graph G:

$$W = \begin{bmatrix} 0 & 2 & 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 1 & 1 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

Graph G



DEGREE MATRIX

Suppose $G = (V, E, W)$ is a connected undirected graph.

Recall $\delta_i = \sum w_{ij}$

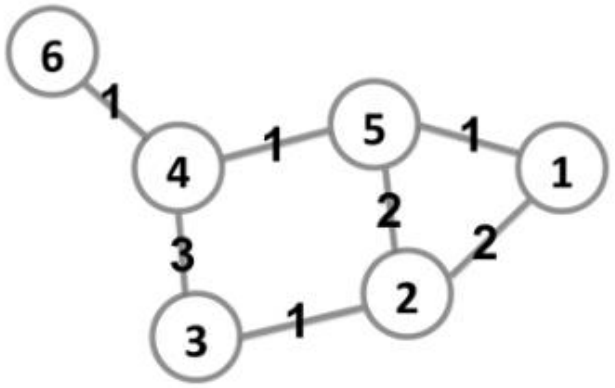
Degree Matrix *D*:

$$D_{ij} = \begin{cases} \delta_i, & i = j \\ 0, & i \neq j \end{cases}$$

Degree Matrix for Graph G:

$$D = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Graph G



LAPLACIAN MATRIX

Suppose $G = (V, E, W)$ is a connected undirected graph.

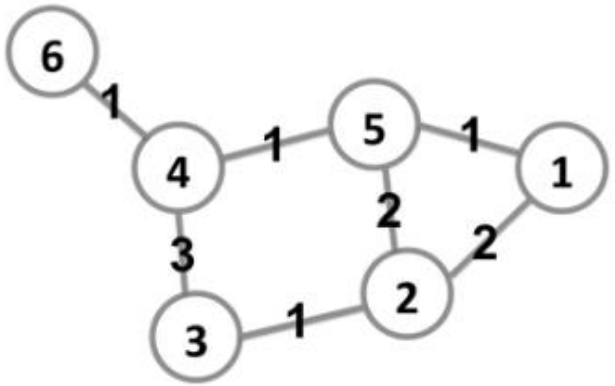
Laplacian Matrix *L* : $L = D - W$

$$W = \begin{bmatrix} 0 & 2 & 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 1 & 1 \\ 1 & 2 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

$$D = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$L = D - W = \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

Graph G



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$$D = \begin{bmatrix} 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$L = \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

Note:

- L is symmetric
- $L\vec{1} = \vec{0}$

Proof:

$L\vec{1}$

$$= \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 - 2 - 1 \\ -2 + 5 - 1 - 2 \\ -1 + 4 - 3 \\ -3 + 5 - 1 - 1 \\ -1 - 2 - 1 + 4 \\ -1 + 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

We can rewrite as $L\vec{1} = \vec{0} = 0 \cdot \vec{1}$

$$L\vec{1} = \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = 0 \cdot \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Definition. Let A be an $n \times n$ matrix.

If $Av = \lambda v$ for a nonzero vector v, we say that λ is the **eigenvalue** for vector v, and that v is an **eigenvector** for λ.

$\therefore$ 0 is an eigenvalue of L, and $\vec{1}$ is the corresponding eigenvector

PROPERTIES OF THE LAPLACIAN MATRIX

Suppose $L \in \mathbb{R}^{n \times n}$ is the Laplacian matrix for a weighted undirected connected graph with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ and corresponding eigenvectors $\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_n$

L is symmetric positive semi-definite (all the eigenvalues are non-negative): $0 = \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_n$

λ_2 is the smallest nonzero eigenvalue, called the **Fiedler value** (corresponding eigenvector $\vec{x}_2$ is the **Fiedler vector**)

2-WAY GRAPH PARTITIONING

2-WAY GRAPH PARTITIONING: PROBLEM SETTING

A weighted undirected graph $G = (V, E, W)$ can be partitioned into two disjoint sets, V_1 and V_2 where $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \emptyset$, by removing edges connecting the two parts:

$$\mathbf{cut}(V_1, V_2) = \sum_{i \in V_1, j \in V_2} w_{ij}$$

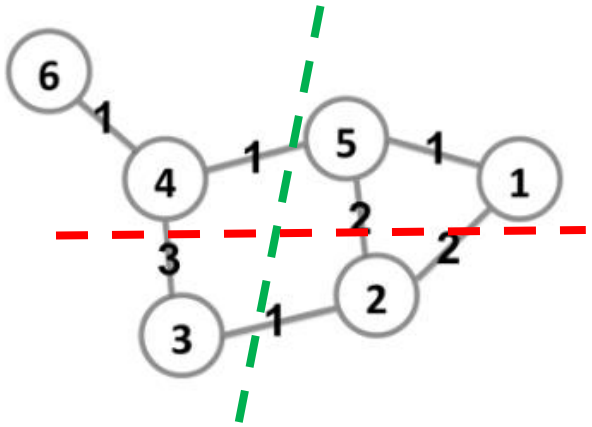
<u>Minimization Problem for $V = V_1 \cup V_2$</u>:

Assume $n = |V|$ is an even number.

min **cut**(V_1, V_2), subject to $|V_1| = |V_2|$

Graph G

$\mathbf{cut}(V_1, V_2) = 2$
$V_1 = \{1,2,5\}$
$V_2 = \{3,4,6\}$



$\mathbf{cut}(V_1, V_2) = 7$
$V_1 = \{1,5,4,6\}$
$V_2 = \{2,3\}$

MINIMIZING THE PARTITION

$$\mathbf{cut}(V_1, V_2) = \sum_{i \in V_1, j \in V_2} w_{ij}$$

<u>Minimization Problem for $V = V_1 \cup V_2$:</u>

Assume $n = |V|$ is an even number.

min $\mathbf{cut}(V_1, V_2)$, subject to $|V_1| = |V_2|$

<u>Equivalent Minimization Problem for $V = V_1 \cup V_2$:</u>

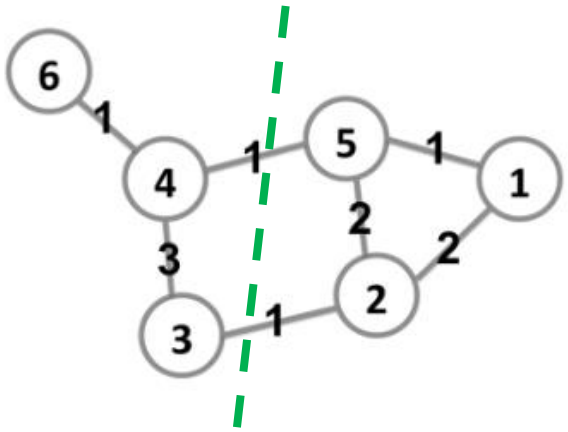
$\min\limits_{\vec{0} \neq \vec{x} \in \mathbb{R}^n} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}$, subject to $x_i = \pm \frac{1}{\sqrt{n}}$ and $|V_1| = |V_2|$

Note: $\left|\left|\vec{x}\right|\right| = 1$ and $\vec{x}^T \vec{1} = 0$

<u>Approximate Minimization Problem for $V = V_1 \cup V_2$:</u>

$\min\limits_{\vec{0} \neq \vec{x} \in \mathbb{R}^n} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}$, subject to $\left|\left|\vec{x}\right|\right| = 1,\ \vec{x}^T \vec{1} = 0$

Graph G



Approximate Minimization Problem for $V = V_1 \cup V_2$:

$$\min_{\vec{0} \neq \vec{x} \in \mathbb{R}^n} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}, \text{ subject to } ||\vec{x}|| = 1,\ \vec{x} \perp \vec{1}$$

Min-Max Theorem

Let L be a symmetric matrix $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ with corresponding eigenvectors $\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_n$, then

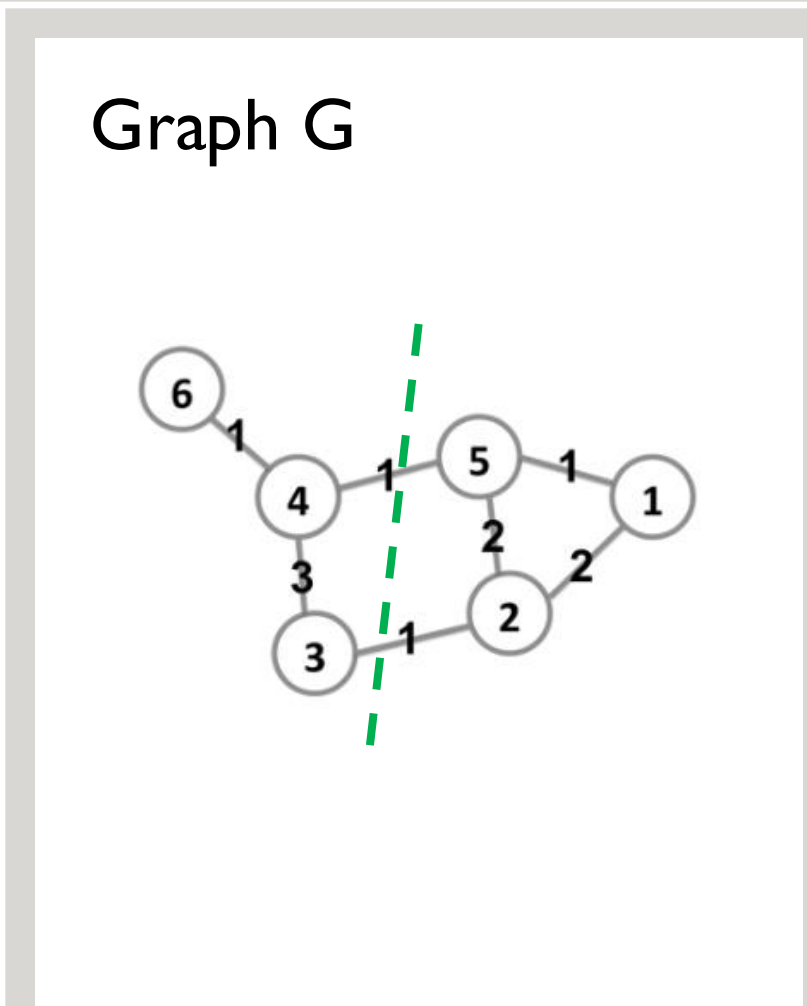
$$\lambda_k = \min_{\vec{x} \perp \vec{x}_1, \ldots, \vec{x}_{k-1}} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}$$

$$\vec{x}_k = \operatorname*{argmin}_{\vec{x} \perp \vec{x}_1, \ldots, \vec{x}_{k-1}} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}$$

When k=2:

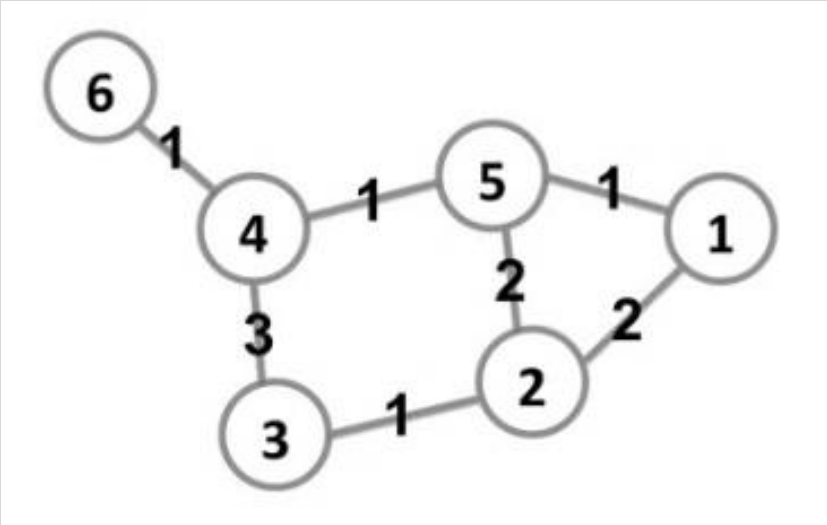
$$\vec{x}_2 = \operatorname*{argmin}_{\vec{x} \perp \vec{x}_1} \frac{\vec{x}^T L \vec{x}}{\vec{x}^T \vec{x}}$$

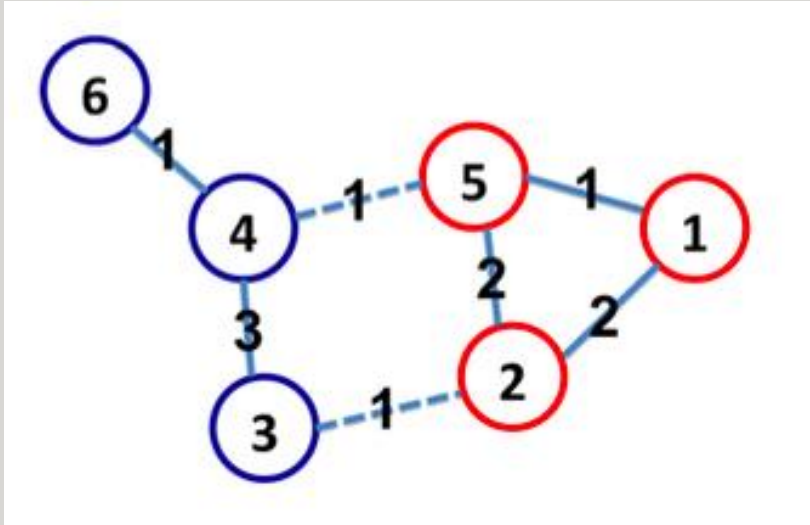
The minimizer is the Fiedler vector $\vec{x}_2$



Compute the Fiedler vector $\vec{x}_2$ corresponding to λ_2 of L

$$L = \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}, \qquad \vec{x}_2 = \begin{bmatrix} -0.41 \\ -0.31 \\ 0.05 \\ 0.16 \\ -0.28 \\ 0.79 \end{bmatrix}$$



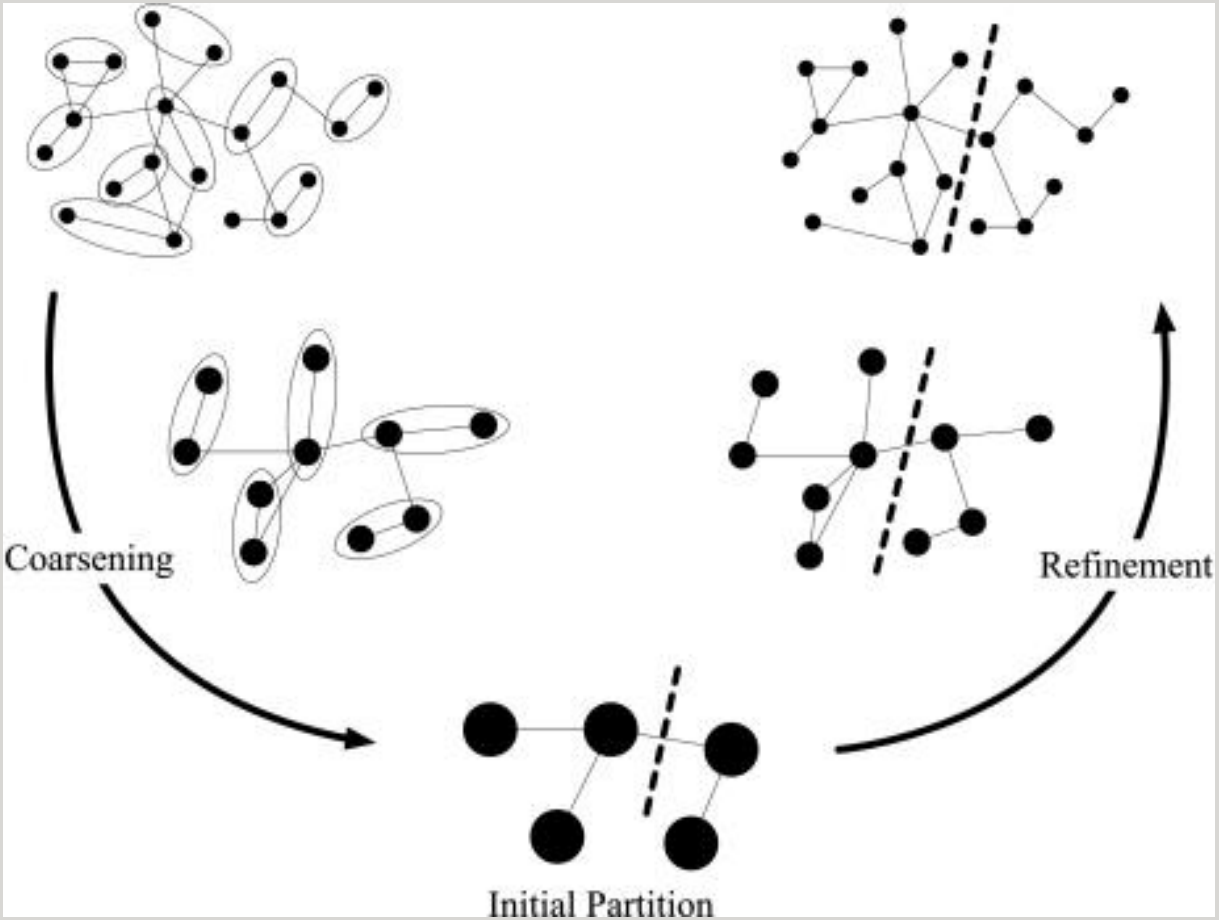


ALGEBRAIC MULTIGRID METHOD (WITH HEAVY EDGE COARSENING)

OUR MOTIVATION

Partitioning a large graph is time-consuming and expensive.

Algebraic Multigrid (AMG) Method: coarsen the graph, solve at a faster speed / lower cost, and refine to our original graph.



COARSENING GRAPH G

Step 1: Group Nodes

$$\{1,2,5\} \rightarrow A$$
$$\{3,4,6\} \rightarrow B$$

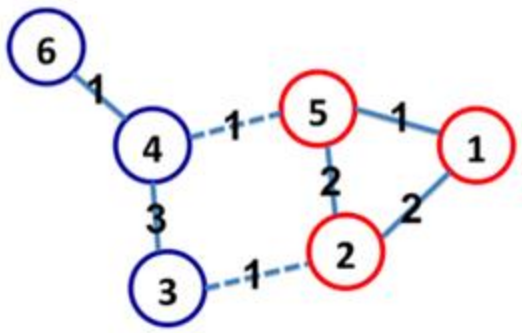
Step 2: Build New Laplacian

$$L(G) = \begin{bmatrix} 3 & -2 & 0 & 0 & -1 & 0 \\ -2 & 5 & -1 & 0 & -2 & 0 \\ 0 & -1 & 4 & -3 & 0 & 0 \\ 0 & 0 & -3 & 5 & -1 & -1 \\ -1 & -2 & 0 & -1 & 4 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

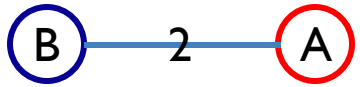
Restriction Matrix $I = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$L(G_c) = I^T L(G) I = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

Original Graph G



Coarsened Graph G_c



ALGEBRAIC MULTIGRID METHOD (WITH HEAVY EDGE COARSENING)

Algorithm 2.1 Multilevel Cascadic Eigensolver

Input: graph Laplacian matrix $L^0 \in \mathbb{R}^{n_0 \times n_0}$
Output: approximate Fiedler vector $\tilde{y}^{(0)}$

Step 1: Setup Phase
set $i = 0$
while $n_i > 25$ **do**
  $I_i^{i+1} \leftarrow \mathrm{HEC}(L^i)$
  $L^{i+1} = I_i^{i+1} L^i (I_i^{i+1})^T$
  $i = i + 1$
end while
$J \leftarrow i$

Step 2: Coarsest Level Solving Phase
$\tilde{y}^{(J)} \leftarrow \mathrm{PI}(L^J, randn(n_J))$

Step 3: Cascadic Refinement Phase
for $j = J - 1$ **to** 0 **do**
  $\hat{y}^{(j)} = (I_i^{i+1})^T \tilde{y}^{(j+1)}$
  $\tilde{y}^{(j)} \leftarrow \mathrm{PI}(L^j, \hat{y}^{(j)})$
end for

Step 1: Setup Phase

Heavy Edge Coarsening – map each vertex to its heaviest edge

Coarsen graph from G_0 to $G_1, G_2, \ldots, G_J$

Step 2: Coarsest Level Solving Phase

Solve the Fiedler vector for coarse graph G_J by applying power iteration to

$$\mathrm{B} = g * \mathrm{eye(n)} - \frac{g}{n}\overrightarrow{1}\overrightarrow{1}^{\mathrm{T}} - L$$

where $g = ||L||_1$

Step 3: Cascadic Refinement Phase

Extend the Fiedler vector from coarse graph G_J to finer graph G_0

[1] Urschel, John C., et al. "A cascadic multigrid algorithm for computing the Fiedler vector of graph Laplacians." *Journal of Computational Mathematics* (2015): 209-226.

APPLYING AMG TO OUR IMAGE

Input: Digital Image

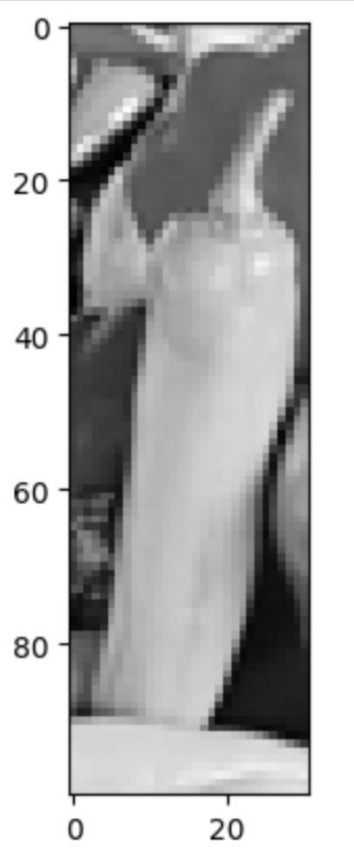
-Compute weights: $w_{ij} = \exp\left(-\frac{\left\|p_i - p_j\right\|^2}{\sigma^2}\right)$ with $\sigma = 5$

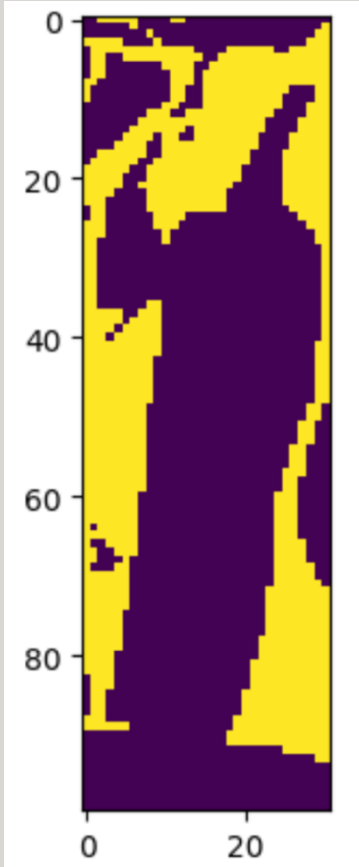
-Apply AMG with HEC to compute Fiedler vector $\vec{x}$

Output: Result Image

If $x_i \geq 0$, set to purple.

If $x_i < 0$, set to yellow.





CONCLUSION

- An image can be represented as a weighted graph
- Graphs can be translated to matrices and solved directly
- Apply AMG method with HEC to find eigenvector efficiently
- Use eigenvector for image segmentation

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John C Urschel, Jinchao Xu, Xiaozhe Hu, and Ludmil T Zikatanov.
A cascadic multigrid algorithm for computing the fiedler vector of graph laplacians.
Journal of Computational Mathematics, 33(2):209–226, 2015.

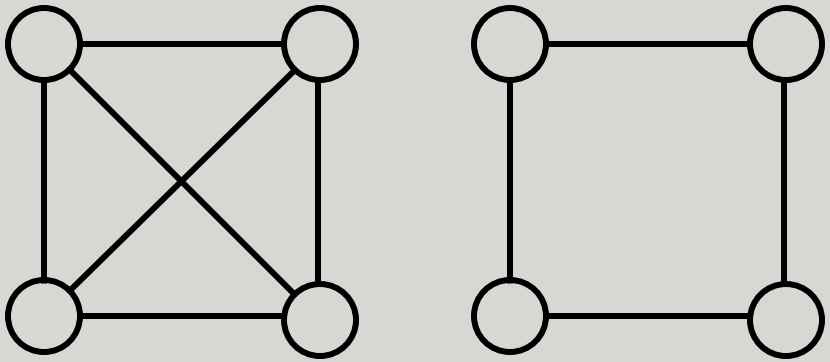
THANK YOU

APPENDIX

Complete Graph (K_n): a simple graph in which there exists an edge between every pair of vertices

Regular Graph: $d(v_i) = r$ for all $v_i \in V$ (r-regular)

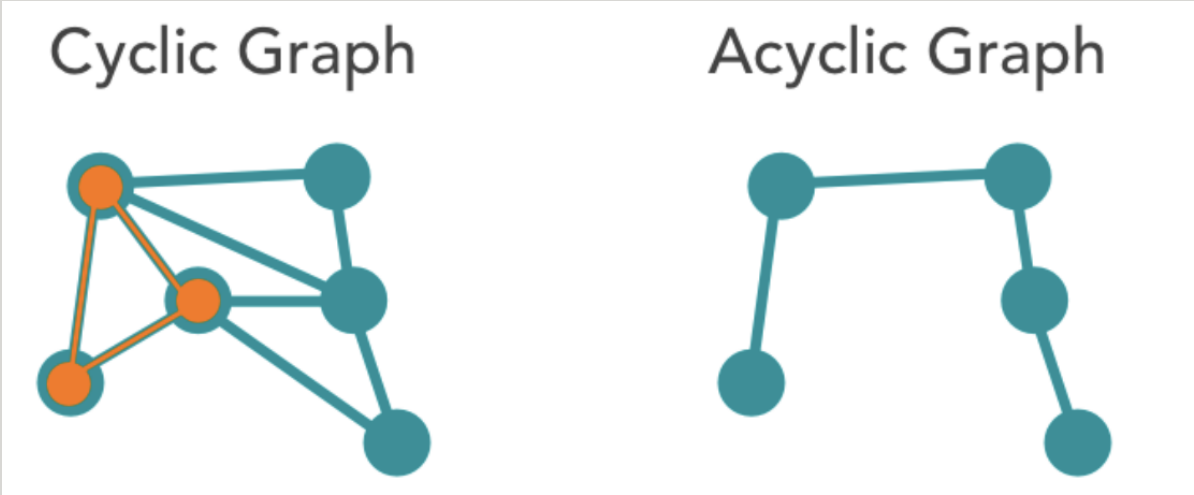
Cycle (C_n): a cycle of n vertices



Empty (Trivial) Graph: a graph with no edges (single vertex)

Consider a sequence of edges $(e_1, e_1, \dots, e_n)$ in a graph G,

- If $e_i = (v_{i-1}, v_i)$ for $i = 1,2, \dots, n$, then the sequence is a **walk**
- If $v_0 = v_n$, a walk is a **closed walk**
- If e_i are distinct, a walk is a **trail**
- If $v_0 = v_n$, a trail is a **closed trail**
- If v_i are distinct, a trail is a **path**
- If $v_0 = v_n$ and $n \geq 3$, a path is a **cycle**
- **Acyclic Graph**: a graph that does not contain a cycle



Connected Vertices: v_i is connected to v_j if there is a path from v_i to v_j

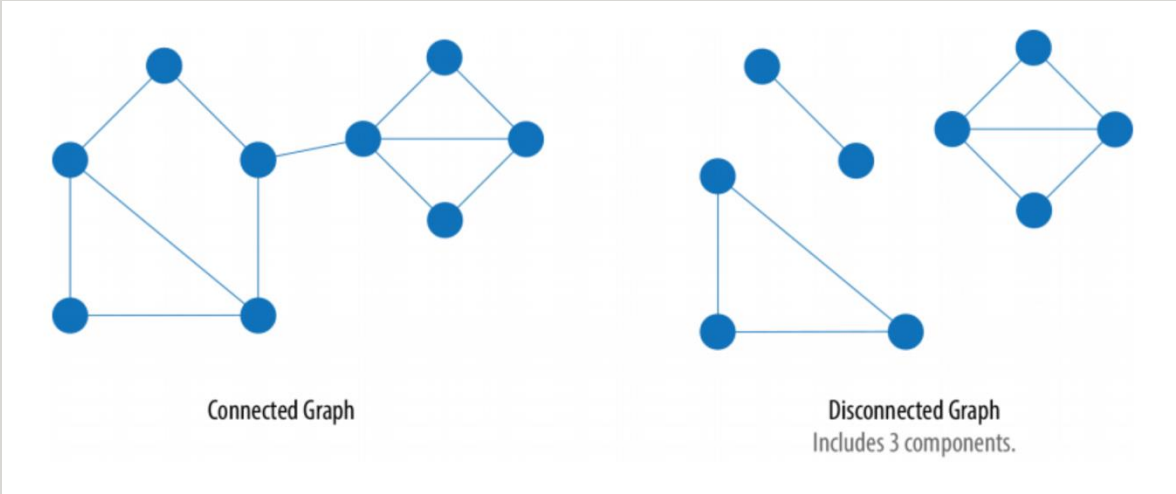
Connected Graph: every two vertices of G are connected

Disconnected Graph: a graph that is not connected

Connected Component: a connected subgraph which is connected to no additional vertices

Lemma: a connected graph with n vertices has at least $n - 1$ edges

Lemma: an acyclic graph with n vertices has at most $n - 1$ edges



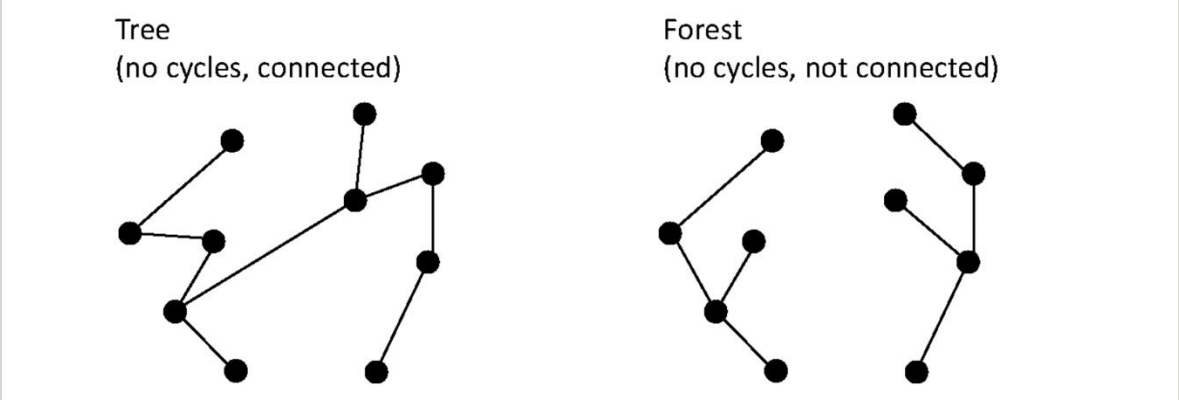
Theorem

Let G be a graph with n vertices, then any two of the following conditions imply the third

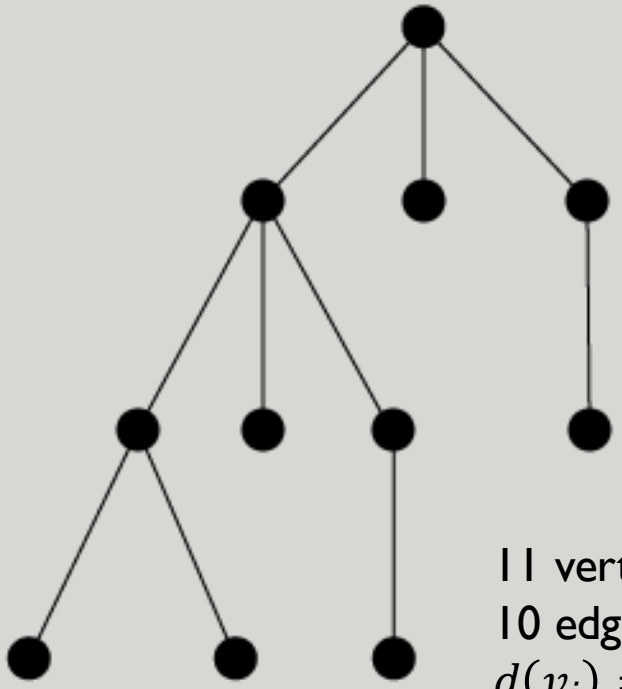
1. G is *connected*
2. G is acyclic
3. G has $n - 1$ *edges*

A graph *T* satisfying all the above conditions is called a **tree**

- **Leaf** v_i: $v_i \in V(T)$ and $d(v_i) = 1$
- **Forest**: a graph whose connected components are trees



Graph *T*



11 vertices
10 edges
$d(v_i) = 1$ for all v_i in *T*

EQUIVALENT MINIMIZATION PROBLEM

Define $\vec{\mathbf{x}} \in \mathbb{R}^n$, where $x_i = \begin{cases} +\dfrac{1}{\sqrt{n}}, & \text{if } i \in \mathcal{V}_1, \\ -\dfrac{1}{\sqrt{n}}, & \text{if } i \in \mathcal{V}_2. \end{cases}$

$$\begin{aligned} \vec{\mathbf{x}}^T L \vec{\mathbf{x}} &= \sum_{i,j \in V} \frac{1}{2} w_{ij} (x_i - x_j)^2 \text{ (next page)} \\ &= \sum_{i \in V_1, j \in V_2} w_{ij} \frac{2}{n} \\ &= \frac{2}{n} \sum_{i \in V_1, j \in V_2} w_{ij} \end{aligned}$$

$$\min \mathbf{cut}(\mathcal{V}_1, \mathcal{V}_2) \iff \min \tfrac{2}{n} \sum_{i \in \mathcal{V}_1, j \in \mathcal{V}_2} w_{ij} = \min \vec{\mathbf{x}}^T L \vec{\mathbf{x}}$$

EQUIVALENT MINIMIZATION PROBLEM

$$
\begin{aligned}
\vec{\mathbf{x}}^T L\vec{\mathbf{x}} &= \sum_{i,j=1}^{n} x_i x_j l_{ij} = \sum_{i=j=1}^{n} l_{ii} x_i^2 + \sum_{i\neq j}^{n} l_{ij} x_i x_j \\
&= \sum_{i=1}^{n} d_i x_i^2 + \sum_{i\neq j}^{n} -w_{ij} x_i x_j \\
&= \sum_{i=1}^{n} (\sum_{j=1}^{n} w_{ij}) x_i^2 - \sum_{i\neq j}^{n} w_{ij} x_i x_j \text{ (by def of } d_i) \\
&= \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} x_i^2 - \sum_{i\neq j}^{n} w_{ij} x_i x_j \\
&= \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} x_i^2 + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} x_i^2 - \sum_{i\neq j}^{n} w_{ij} x_i x_j \\
&= \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} x_i^2 + \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} w_{ij} x_j^2 - \sum_{i\neq j}^{n} w_{ij} x_i x_j \text{ (switch i and j, then use } w_{ij} = w_{ji}) \\
&= \sum_{i,j=1}^{n} \frac{1}{2} w_{ij} (x_i^2 - 2x_i x_j - x_j^2) \\
&= \sum_{i,j=1}^{n} \frac{1}{2} w_{ij} (x_i - x_j)^2
\end{aligned}
$$